

Intro stats with mosaic (lattice version)

Essential R syntax

Names in R are *case sensitive*
 Function and arguments
rflip(10)
 Optional arguments
rflip(10, prob = 0.8)
 Assignment
x <- rflip(10, prob = 0.8)
 Getting help on any function
help(mean)

Loading packages

```
library(mosaic)
```

Arithmetic operations

+	-	*	/	basic operations
^				exponentiation
()				grouping
sqrt(x)				square root
abs(x)				absolute value
log10(x)				logarithm, base 10
log(x)				natural logarithm, base e
exp(x)				exponential function e^x
factorial(k)				$k! = k(k-1) \dots 1$

Logical operators

== is equal to (note double equal sign)
!= is not equal to
< is less than
<= is less than or equal to
> is greater than
>= is greater than or equal to
& **A & B** is **TRUE** if both **A** and **B** are **TRUE**
| **A | B** is **TRUE** if one or both of **A** and **B** are **TRUE**
%in% includes; for example
"C" %in% c("A", "B") is **FALSE**

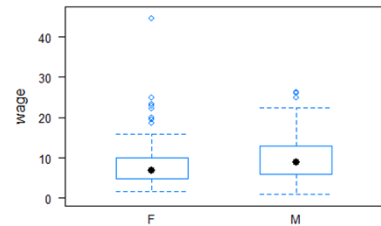
Formula interface

Use for graphics, statistics, inference, and modeling operations.

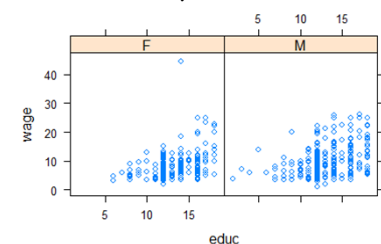
goal(y ~ x, data = mydata)
 Read as "Calculate **goal** for **y** using **mydata** "broken down by" **x**, or "modeled by" **x**.
mean(age ~ sex, data = HELPrct)

For graphics:

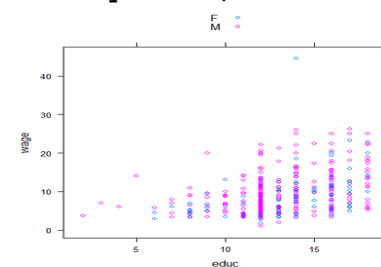
goal(y ~ x | z, groups = w, data = mydata)
y : y-axis variable (*optional*)
x : x-axis variable (*required*)
z : panel-by variable (*optional*)
w : color-by variable (*optional*)
bwplot(wage ~ sex, data = CPS85)



xyplot(wage ~ educ | sex, data = CPS85)



xyplot(wage ~ educ, groups = sex, data = CPS85, auto.key = TRUE)



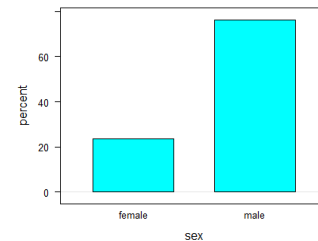
One categorical variable

Counts by category

```
tally(~ sex, data = HELPrct)
```

Percentages by category

```
tally(~ sex, format = "percent", data = HELPrct)
bargraph(~ sex, type = "percent", data = HELPrct)
```



Tests and confidence intervals

Exact test

```
result1 <- binom.test(~ (homeless == "homeless"), data = HELPrct)
```

Approximate test (large samples)

```
result2 <- prop.test(~ (homeless == "homeless"), data = HELPrct)
```

Extract confidence intervals and p -values

```
confint(result1)
pval(result2)
```

Examining data

Print short summary of all variables

```
inspect(HELPrct)
```

Number of rows and columns

```
dim(HELPrct)
nrow(HELPrct)
ncol(HELPrct)
```

Print first rows or last rows

```
head(KidsFeet)
tail(KidsFeet, 10)
```

Names of variables

```
names(HELPrct)
```

One quantitative variable

Make output more readable

```
options(digits = 3)
```

Compute summary statistics

```
mean(~ cesd, data = HELPrct)
```

Other summary statistics work similarly

```
median() iqr() max() min()
fivenum() sd() var() sum()
```

Table of summary statistics

```
favstats(~ cesd, data = HELPrct)
```

Summary statistics by group

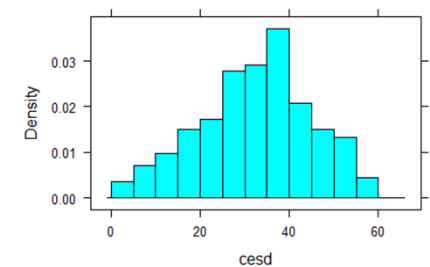
```
favstats(cesd ~ sex, data = HELPrct)
```

Quantiles

```
quantile(~ cesd, data = HELPrct, prob = c(0.25, 0.5, 0.8))
```

Histogram

```
histogram(~ cesd, width = 5, center = 2.5, data = HELPrct)
```



Normal probability plot

```
qqmath(~ cesd, dist = "qnorm", data = HELPrct)
```

Density plot

```
densityplot(~ cesd, data = HELPrct)
```

Dot plot

```
dotPlot(~ cesd, data = HELPrct)
```

One-sample t -test

```
result <- t.test(~ cesd, mu = 34, data = HELPrct)
```

Extract confidence intervals and p -values

```
confint(result)
pval(result)
```

Two categorical variables

Contingency table with margins

```
tally(~ substance + sex,
      margins = TRUE,
      data = HELPrct)
```

Percentages by column

```
tally(~ sex | substance,
      format = "percent",
      data = HELPrct)
```

Mosaic plot

```
mosaicplot(~ substance + sex,
            color = TRUE, data = HELPrct)
```



Chi-square test

```
xchisq.test(~ substance + sex,
             data = HELPrct,
             correct = FALSE)
```

Distributions

Normal distribution function

```
pnorm(13, mean = 10, sd = 2)
```

Normal distribution function with graph

```
xpnorm(1.645, mean = 0, sd = 1)
```

Normal distribution quantiles

```
qnorm(0.95) # mean = 0, sd = 1
```

Normal distribution quantiles with graph

```
xqnorm(0.85, mean = 10, sd = 2)
```

Binomial density function ("size" means n)

```
dbinom(5, size = 8, prob = 0.65)
```

Binomial distribution function

```
pbinom(5, size = 8, prob = 0.65)
```

Central portion of distribution

```
cdist("norm", 0.95)
cdist("t", c(0.90, 0.99), df = 5)
```

Plotting distributions

```
plotDist("binom", size = 8,
         prob = 0.65, xlim = c(-1, 9))
plotDist("norm", mean = 10,
         sd = 2)
```

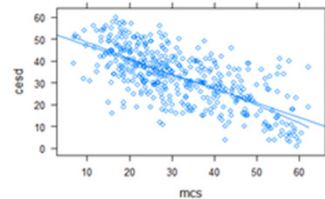
Two quantitative variables

Correlation coefficient

```
cor(cesd ~ mcs, data = HELPrct)
```

Scatterplot with regression line and smooth

```
xyplot(cesd ~ mcs,
       type = c("p", "r", "smooth"),
       data = HELPrct)
```



Simple linear regression

```
cesdmodel <- lm(cesd ~ mcs,
                 data = HELPrct)
msummary(cesdmodel)
```

Prediction

```
lmfunction <- makeFun(cesdmodel)
lmfunction(mcs = 35)
```

Extract useful quantities

```
anova(cesdmodel)
coef(cesdmodel)
confint(cesdmodel)
rsquared(cesdmodel)
```

Diagnostics; plot residuals

```
histogram(~resid(cesdmodel),
          density = TRUE)
qqmath(~resid(cesdmodel))
```

Diagnostics; plot residuals vs. fitted

```
xyplot(resid(cesdmodel) ~
       fitted(cesdmodel),
       type = c("p", "smooth", "r"))
```

Categorical response, quantitative predictor

Logistic regression

```
logit_mod <-
  glm(homeless ~ age + female,
      family = binomial, data = HELPrct)
msummary(logitmod)
```

Odds ratios and confidence intervals

```
exp(coef(logit_mod))
exp(confint(logit_mod))
```

Data management

From dplyr package

Drop or reorder variables

```
select()
```

Create new variables from existing ones

```
mutate()
```

Retain specific rows from data

```
filter()
```

Sort data rows

```
arrange()
```

Compute summary statistics by group

```
group_by()
summarize()
```

Merge data tables

```
left_join()
inner_join()
```

Importing data

Import file from computer or URL

```
MustangPrice <-
  read.file("C:/MustangPrice.csv")
# NOTE: R uses forward slashes!
Dome <-
  read.file("http://www.mosaic-
  web.org/go/datasets/Dome.csv")
```

Randomization and simulation

Fix random number sequence

```
set.seed(42)
```

Tossing coins

```
rflip(10) # default prob is 0.5
```

Do something repeatedly

```
do(5) * rflip(10, prob = 0.75)
```

Draw a simple random sample

```
sample(LETTERS, 10)
deal(Cards, 5) # poker hand
```

Resample with replacement

```
Small <- sample(KidsFeet, 10)
resample(Small)
```

Random permutation (shuffling)

```
shuffle(Cards)
```

Random values from distributions

```
rbinom(5, size = 10, prob = 0.7)
rnorm(5, mean = 10, sd = 2)
```

Quantitative response, categorical predictor

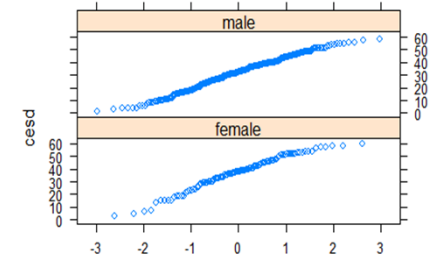
Two-level predictor: two-sample t test

Numeric summaries

```
favstats(~cesd | sex,
        data = HELPrct)
```

Comparative normal probability plot

```
qqmath(~cesd | sex, data = HELPrct,
       layout = c(1, 2)) # also bwplot
```



Dotplot for smaller samples

```
xyplot(sex ~ length, alpha = 0.6,
       cex = 1.4, data = KidsFeet)
```

Two-sample t -test and confidence interval

```
result <- t.test(cesd ~ sex,
                 var.equal = FALSE, data = HELPrct)
confint(result)
```

More than two levels: Analysis of variance

Numeric summaries

```
favstats(cesd ~ substance,
        data = HELPrct)
```

Graphic summaries

```
bwplot(cesd ~ substance, pch = "|",
       data = HELPrct)
```

Fitt and summarize model

```
modsubstance <- lm(cesd ~ substance,
                  data = HELPrct)
anova(modsubstance)
```

Which differences are significant?

```
pairwise <- TukeyHSD(modsubstance)
mplot(pairwise)
```

95% family-wise confidence level

